

On regularization ambiguities in fermionic models

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Different regularization schemes based on the heat-kernel method using as samples the chiral Schwinger model and the Thirring model are discussed. As it happens with non-Abelian anomalies, it is shown that different physical contexts require different specifications of the fermionic generating functional. Also discussed is the introduction of an arbitrary parameter through the regularization and how this affects the resulting quantum theory.

I. INTRODUCTION

Quantization of gauge theories with Weyl fermions has recently received a lot of attention, particularly after the results of Jackiw and Rajaraman¹ on the consistency of the chiral Schwinger model and those of Faddeev and Shatashvili² on the modification of the canonical quantization by addition of new degrees of freedom through a Wess–Zumino action.

Prompted by these important developments, many investigations^{3–11} analyzed the possibility of quantization of anomalous theories, usually considered as inconsistent—nonunitary and nonrenormalizable—if cancellation of anomalies was not carefully contrived.

Analysis of potentially anomalous theories requires the development of regularization procedures that take into account the peculiarities of chiral fermions. In the path-integral approach, where most of the above mentioned advances were made, calculation of anomalies depends on a precise understanding of the functional integration measure and on how it changes under the original classical symmetry of the theory.^{12,13} In this approach one defines a Jacobian J associated to this symmetry transformation (known as the Fujikawa Jacobian¹²) as a ratio of fermion determinants. If this Jacobian is not trivial, the associated Noether current is anomalous.

The crucial point is that each determinant appearing in J needs a regularization (for example, for Dirac or Weyl fermions the Dirac operators appearing in J are unbounded and hence determinants are ill-defined quantities).

Typically, one regulates J by inserting $\exp(-R/M^2)$ (Refs. 12 and 13) with R some positive definite operator called the regularization operator (RO); then one performs all computations keeping M^2 fixed and finally one takes the limit $M^2 \rightarrow \infty$ at the end of the calculation. This method is usually known as the heat-kernel method. It is important to note that the choice of RO determines whether the Jacobian is trivial or not.

If one has more than one classical symmetry it may happen that the measure and the RO cannot be made invariant under all of them. It is then a matter of physical prejudice as to which symmetries one maintains at the quantum level and which symmetry becomes anomalous. Alternatively, it may happen that there is no choice of RO respecting the original symmetry of the theory (this last case corresponds to the

case of chiral models). In any case, anomalies arise.

For gauge theories with Dirac fermions, the fermionic Lagrangian dictates the choice of RO. Indeed, taking $R = D^2$ (Refs. 12 and 13), with $D = i\not{\partial} + g\not{A}$, the (Hermitian) Dirac operator appearing in the Lagrangian, gauge invariance, the fundamental symmetry to be preserved, is ensured. With this choice classical chiral symmetry is sacrificed at the quantum level. The resulting anomaly in the axial current obtained using the heat-kernel approach, coincides with the perturbative results¹⁴ as well as with the ones obtained using the ζ -function method. (In fact, for Hermitian D one can show the equivalence between the heat-kernel method and ζ -function method.¹³)

If A_μ does not represent a gauge field (for example, when it is introduced as an auxiliary field in purely fermionic models like the Thirring model) then it is not compulsory to make the choice $R = D^2$. More general RO can in fact be introduced¹⁵ and in this form more general results obtained (as it happens in the Thirring model case, when a one-parameter family of solutions is known to exist¹⁶).

Gauge theories with Weyl fermions are another example in which regularization ambiguities arise. This is due to the following peculiarities of chiral models.^{17,18}

Primum: The Dirac operator appearing in the Lagrangian, $D = \not{D}(1 - \gamma_5)/2$ (we take left-handed fermions for definiteness), does not define an eigenvalue problem (it maps negative chirality spinors into positive chirality ones).

Secundum: A modified Dirac operator $\tilde{D}$, constructed by addition of a free right-handed sector so as to define an eigenvalue problem, is neither gauge-covariant nor Hermitian. Either analytic continuation of $\tilde{D}$ (Ref. 18) or use of $R = (\tilde{D}\tilde{D}^\dagger + \tilde{D}^\dagger\tilde{D})/2$ (Ref. 10) breaks gauge invariance at the quantum level.

Tertium: A family of RO depending on certain parameters are in fact admissible.^{19–26} This leads to regularization ambiguities that can be exploited to render the quantum theory consistent (at least in two space-time dimensions, as first shown in Ref. 1).

We address these regularization problems in the present work using the chiral-Schwinger model (CSM) and the Thirring model (TM) as examples. Solutions of these two-dimensional models are known and hence their path-integral treatment can be used to get insight into the more general problem of regulating Jacobians and fermion determinants.

The paper is organized as follows. In Sec. II, after briefly

defining the models (II A) we discuss the change in the fermionic measure under chiral and gauge transformations (Sec. II B). We then discuss the regularization dependence of Jacobians and fermion determinants for the CSM and TM in Sec. II C.

From these results we evaluate in Sec. III fermion currents, anomalies, and current-current commutators showing how different regularization schemes affect the results. We give in Sec. IV a summary of our results and conclusions leaving for an Appendix details of the calculations.

II. FERMION DETERMINANTS USING DIFFERENT REGULARIZATIONS

A. The models

We shall consider two two-dimensional fermionic models where regularization ambiguities may arise: the Thirring model and the chiral Schwinger model. The first one is defined by the (Euclidean) Lagrangian

$$\mathcal{L} = \bar{\psi} i \not{\partial} \psi - (g^2/2) (\bar{\psi} \gamma_\mu \psi)^2. \quad (2.1)$$

Our conventions for matrices are

$$\gamma_\mu \gamma_\nu + \gamma_\nu \gamma_\mu = 2\delta_{\mu\nu},$$

$$\gamma_5 = i\gamma_0\gamma_1,$$

$$\varepsilon_{01} = -\varepsilon_{10} = 1.$$

The generating functional for this model is

$$\mathcal{Z}_{\text{TM}}[S_\mu] = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp\left(-\int d^2x (\mathcal{L} + \bar{\psi} S_\mu \psi)\right), \quad (2.2)$$

where $S_\mu(x)$ is an external source.

Using the identity²⁷

$$\begin{aligned} \exp\left(\frac{g^2}{2} \int (\bar{\psi} \gamma_\mu \psi)^2 d^2x\right) \\ = \int \mathcal{D}A_\mu \exp\left(-\int \left[\frac{1}{2} A_\mu^2 + g \bar{\psi} A_\mu \psi\right] d^2x\right), \end{aligned} \quad (2.3)$$

the generating functional can be rewritten in the form

$$\begin{aligned} \mathcal{Z}_{\text{TM}}[S_\mu] = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \mathcal{D}A_\mu \exp\left(-\int \left[\frac{1}{2} A_\mu^2 \right. \right. \\ \left. \left. + \bar{\psi} S_\mu \psi + \bar{\psi} \not{D}[A] \psi\right] d^2x\right), \end{aligned} \quad (2.4)$$

with

$$\not{D}[A] = i\not{\partial} + g\not{A}. \quad (2.5)$$

Note that the auxiliary field A_μ represents two degrees of freedom: due to the presence of the A_μ^2 term in (2.4), no gauge fixing has to be implemented. Then, when regularizing the fermionic path integral (i.e., the fermionic determinant), gauge invariance cannot be invoked to select the regulator.

A similar situation arises in the quantization of the chiral Schwinger model.¹ Indeed, the fermionic Lagrangian for this gauge theory coupled to left-handed fermions is

$$\mathcal{L}_{\text{CSM}} = \bar{\psi}(i\not{\partial} + g\not{A})((1 - \gamma_5)/2)\psi = \bar{\psi} \not{D}[A] \psi. \quad (2.6)$$

The fermionic part of the generating functional reads

$$\begin{aligned} \mathcal{Z}_{\text{CSM}} &= \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp\left(-\int \mathcal{L}_{\text{CSM}} d^2x\right) \\ &= \det D_{\text{CSM}}[A]. \end{aligned} \quad (2.7)$$

Now, since the Dirac operator for chiral fermions ($D_{\text{CSM}}[A]$) maps negative chirality spinors into positive chirality ones, it does not define a correct eigenvalue problem. Usually one overcomes this difficulty by introducing right-handed free fermions,^{17,18}

$$\begin{aligned} \tilde{D}[A] &= D_{\text{CSM}}[A] + i\not{\partial}((1 + \gamma_5)/2) \\ &= i\not{\partial} + g\not{A}((1 - \gamma_5)/2) \end{aligned} \quad (2.8)$$

and defining

$$\det D_{\text{CSM}}[A] = \det \tilde{D}[A] \big|_{\text{Reg}}. \quad (2.9)$$

We have indicated with the subscript “Reg” that some regularization has to be adopted since $\tilde{D}[A]$ is an unbounded operator. Now, the addition of right-handed fermions (which can be justified by several arguments¹⁷) solves the chirality-flip problem but introduces an ambiguity. Indeed, $\tilde{D}[A]$ is not a covariant derivative and hence there is no reason to expect

$$\det \tilde{D}[A] \big|_{\text{Reg}} = \det \tilde{D}[A^g] \big|_{\text{Reg}}$$

with A_μ^g the gauge transformed of A_μ . As in the Thirring model case, there is then no reason to use a gauge-invariant regularization prescription.

B. Jacobians

For notation brevity we shall call $D[A]$ either $\not{D}[A]$ defined in (2.5) for the Thirring model or $\tilde{D}[A]$, defined in (2.8) for the chiral Schwinger model.

We shall evaluate the change in the fermionic determinant (i.e., the Fujikawa Jacobian¹²) associated with the following change of variables:

$$\psi = u_t \psi', \quad \bar{\psi} = \bar{\psi}' \tilde{u}_t. \quad (2.10)$$

For the Thirring model u_t and $\tilde{u}_t$ are given by

$$\begin{aligned} u_t &= e^{[\gamma_5 \phi(x) + i\eta(x)]t}, \\ \tilde{u}_t &= e^{[\gamma_5 \phi(x) - i\eta(x)]t}, \end{aligned} \quad (2.11)$$

with η (ϕ) a scalar (pseudoscalar) field and t a real parameter.

For the CSM, since the fermions are left-handed, transformation²⁻¹¹ can be simplified to¹⁰

$$\begin{aligned} u_t &= \exp[+((1 - \gamma_5)/2)(\phi - i\eta)t], \\ \tilde{u}_t &= \exp[-((1 + \gamma_5)/2)(\phi - i\eta)t]. \end{aligned} \quad (2.12)$$

Transformations (2.10) and (2.12) are chosen so that for $t = 1$ the corresponding Lagrangians become free. Indeed, if we relate ϕ and η to the vector field A_μ through the identity

$$A_\mu = -(1/g)(\epsilon_{\mu\nu} \partial_\nu \phi - \partial_\mu \eta), \quad (2.13)$$

then, under transformations (2.10)–(2.12) the fermion Lagrangians become

$$\mathcal{L} \rightarrow \mathcal{L}' = \bar{\psi}' \not{D}_t \psi', \quad (2.14)$$

with

$$D_t = D[(1-t)A] \quad (2.15)$$

and hence for $t = 1$ one has $D_1 = i\partial$.

The Fujikawa Jacobian $J(t)$ associated with transformations (2.10)–(2.12) relates the determinants of operators $D_t[A]$ and $D[A]$

$$\det D[A] = J(t) \det D_t[A]. \quad (2.16)$$

Since both determinants require a regularization, the result for $J(t)$ necessarily depends on this regularization. Here $J(t)$ can be evaluated from the following differential equation¹³ [derived from (2.16)]:

$$-\frac{d}{dt} \log J(t) = \frac{d}{dt} \log \det D_t \Big|_{\text{Reg}} \equiv \omega'(t) \quad (2.17)$$

[a regularization is understood in (2.17)].

In particular, the fermion determinant in the lhs of (2.16) can be evaluated from the knowledge of $J(1)$,

$$\det D[A] = J(1) \det i\partial, \quad (2.18)$$

with

$$J(1) = \exp\left(-\int_0^1 \omega'(t) dt\right). \quad (2.19)$$

We evaluate $\omega'(t)$ using

$$\omega'(t) = \lim_{\Delta t \rightarrow 0} \frac{\text{tr} \log D_t^{-1} D_{t+\Delta t}}{\Delta t}. \quad (2.20)$$

For the TM ($D_t = \not{D}_t$) one obtains

$$\omega'_{\text{TM}}(t) = \text{tr} \not{D}_t^{-1} [(\gamma_5 \phi - i\eta) \not{D}_t + \not{D}_t (\gamma_5 \phi + i\eta)]. \quad (2.21)$$

Analogously, for the CSM ($D_t = \tilde{D}_t$) one has

$$\omega'_{\text{CSM}}(t) = \text{tr} [D_t^{-1} (\tilde{D}_t((1-\gamma_5)/2)(\phi - i\eta) - ((1+\gamma_5)/2)(\phi - i\eta)\tilde{D}_t)]. \quad (2.22)$$

As we stated above, $\omega'(t)$ needs a regularization that can in principle destroy the cyclic property of the trace as first observed in Ref. 25. This is a very important point: if the cyclic property is assumed, (2.21) and (2.22) simplify to

$$\omega'_{\text{TM}} = 2 \text{tr} \gamma_5 \phi, \quad (2.23)$$

$$\omega'_{\text{CSM}} = \text{tr} \gamma_5 (\phi - i\eta). \quad (2.24)$$

We shall discuss this point in more details below.

C. Regularization dependence of Jacobians and determinants

For gauge invariant theories, the regularization prescription is chosen so as to respect gauge invariance. One can prove that the ζ -function method, the heat-kernel one [with $\exp(-\not{D}^2/M^2)$ as regulator], etc., lead to the same gauge-invariant answer for the fermionic determinant. Moreover, the cyclic property of the trace holds. (This can be seen very simply in the heat-kernel approach, since $[\exp(-D_t^2/M^2), D_t] = 0$.)

We then have, for gauge-invariant models, for a transformation like (2.10),¹³

$$\omega'(t) = \lim_{M^2 \rightarrow \infty} 2 \text{tr} \gamma_5 \phi e^{-R/M^2} = -\frac{d}{dt} \frac{d\zeta}{ds}(s, D_t) \Big|_{s=0}, \quad (2.25)$$

provided the regulating operator (RO) R in the heat-kernel approach coincides with D_t . As we stated above gauge invariance cannot be invoked in regularizing the Thirring or the CS models. Then, there is no particular reason to use the ζ -function method or the heat-kernel one with $R = D_t$. In fact, more general prescriptions (compatible with Lorentz invariance) can be implemented and we address this point now.

We consider here the heat-kernel approach (for the ζ -function method, see Ref. 24). We shall use as regulating operator the following one^{19–23}:

$$R = [D_t^a]^2 = [i\partial + g(1-t)A_+ + ag(1-t)A_-]^2, \quad (2.26)$$

with

$$A_{\pm} = A((1 \mp \gamma_5)/2). \quad (2.27)$$

In (2.25) a is a parameter related to the one introduced by Jackiw and Rajaraman¹ to take care of regularization ambiguities in the CSM. In the present analysis, its presence gives different weights to the A_+ and A_- components which, for the TM and the CSM correspond to independent degrees of freedom. Another way of understanding the presence of this undetermined parameter is the following. In computing $\log \det(i\partial_+ + gA_+)$ [$\log \det(i\partial_- + gA_-)$] one gets in principle a result $F[A_+]$ ($F[A_-]$) depending only on A_+ (A_-). The complete determinant $F[A_\mu] = \det(i\partial + gA)$ including both right-handed and left-handed parts has necessarily an arbitrariness

$$F[A_\mu] = F[A_+] + F[A_-] + C \int A_\mu^2 d^2x. \quad (2.28)$$

The arbitrary constant can be fixed only on gauge-invariance grounds. One can easily obtain a relation like (2.28) using D_t^a as RO.^{19,23}

Now, since D_t^a is not Hermitian, two possibilities arise when using the heat-kernel approach in order to have a positive definite R (Refs. 13, 28, and 29).

$$\text{Analytic continuation } \gamma_5 \rightarrow i\gamma_5 \text{ in } R_A = D_t^{a^2} \quad (2.29a)$$

or

$$R_B = (D_t^a D_t^{a^+} + D_t^{a^+} D_t^a)/2. \quad (2.29b)$$

The alternative (2.29a) has been shown¹³ to lead to the same result as the ζ function (which is defined even if D is not Hermitian). The alternative (2.29b) was proposed by Fujikawa²⁹ in his analysis of covariant and consistent anomalies.

Now, in both cases,

$$[e^{-R/M^2}; D_t] \neq 0,$$

and then, the regulated forms of (2.21) and (2.22) and (2.23) and (2.24) are not the same. This can be interpreted as if the cyclic property of traces in (2.21) and (2.22) does not hold.²⁵ We shall examine each possibility separately.

(I) One starts from (2.23) and (2.24) (RegI) obtained using cyclic property of the trace. Inserting the RO one gets

$$\omega'_{\text{TM}}(t) \Big|_{\text{RegI}} = \lim_{M^2 \rightarrow \infty} 2 \text{tr} \gamma_5 \phi e^{-R/M^2}, \quad (2.30)$$

$$\omega'_{\text{CSM}}(t) \Big|_{\text{RegI}} = \lim_{M^2 \rightarrow \infty} \text{tr} \gamma_5 (\phi - i\eta) e^{-R/M^2}. \quad (2.31)$$

(II) One starts from (2.21) and (2.22) (RegII), getting

$$\omega'_{\text{TM}}(t) \big|_{\text{RegII}} = \lim_{M^2 \rightarrow \infty} \text{tr} \, \mathcal{D}_t^{-1} [(\gamma_5 \phi - i\eta) \mathcal{D}_t + \mathcal{D}_t (\gamma_5 \phi + i\eta)] e^{-R/M^2}, \quad (2.32)$$

$$\omega'_{\text{CSM}}(t) \big|_{\text{RegII}} = \lim_{M^2 \rightarrow \infty} \text{tr} \, \mathcal{D}_t^{-1} \left[\tilde{\mathcal{D}}_t \left(\frac{1 - \gamma_5}{2} \right) (\phi - i\eta) - \left(\frac{1 + \gamma_5}{2} \right) (\phi - i\eta) \tilde{\mathcal{D}}_t \right] e^{-R/M^2}. \quad (2.33)$$

We shall consider the two above-mentioned choices for R , namely,

$$\begin{aligned} R_A &= D_t^2, \\ R_B &= (D_t D_t^\dagger + D_t^\dagger D_t)/2. \end{aligned} \quad (2.34)$$

We have evaluated $\omega'_{\text{TM}}(t)$ and $\omega'_{\text{CSM}}(t)$ using regularization RegI and RegII, and regulating operator R_A and R_B . We denote these choices in the form RegIA, RegIB, etc. Details of the calculations are given in the Appendix. The answer is

$$\omega'_{\text{TM}} \big|_{\text{RegIA}} = (1-t) \left[\frac{g}{2\pi} (1+a) \int \phi \epsilon_{\mu\nu} \partial_\mu A_\nu d^2x + \frac{i}{2\pi} g(a-1) \int \phi \partial_\mu A_\mu d^2x \right],$$

$$\omega'_{\text{TM}} \big|_{\text{RegIB}} = (1-t) \frac{g}{2\pi} (1+a) \int \phi \epsilon_{\mu\nu} \partial_\mu A_\nu d^2x,$$

$$\omega'_{\text{TM}} \big|_{\text{RegIIA}} = (1-t) \left[\frac{g}{\pi} \int \phi \epsilon_{\mu\nu} \partial_\mu A_\nu d^2x + \frac{g^2}{2\pi} (a-1) \times \int \phi \partial_\mu A_\nu (i\delta_{\mu\nu} + \epsilon_{\mu\nu}) d^2x \right],$$

$$\omega'_{\text{TM}} \big|_{\text{RegIIB}} = (1-t) \frac{g}{2\pi} (1+a) \int \phi \epsilon_{\mu\nu} \partial_\mu A_\nu d^2x,$$

$$\begin{aligned} \omega'_{\text{CSM}} \big|_{\text{RegIA}} &= (1-t) \left[\frac{g}{2\pi} \int (\phi - i\eta) \epsilon_{\mu\nu} \partial_\mu A_\nu d^2x \right. \\ &\quad \left. + \frac{ig}{4\pi} (a-1) \int (\phi - i\eta) \partial_\mu A_\nu (\delta_{\mu\nu} - i\epsilon_{\mu\nu}) d^2x \right], \end{aligned}$$

$$\omega'_{\text{CSM}} \big|_{\text{RegIB}} = (1-t) \frac{g}{2\pi} (1+a) \times \int (\phi - i\eta) \epsilon_{\mu\nu} \partial_\mu A_\nu d^2x,$$

$$\omega'_{\text{CSM}} \big|_{\text{RegIIA}} = (1-t) \frac{g}{4\pi} (1+a) \times \int (\phi - i\eta) \partial_\mu A_\nu (\epsilon_{\mu\nu} - i\delta_{\mu\nu}) d^2x,$$

$$\omega'_{\text{CSM}} \big|_{\text{RegIIB}} = (1-t) \frac{g}{4\pi} (1+a) \times \int (\phi - i\eta) \partial_\mu A_\nu (\epsilon_{\mu\nu} - i\delta_{\mu\nu}) d^2x. \quad (2.35)$$

We shall see in the next section how all these regularization-dependent results for ω' affect the anomaly in fermion currents, current-current commutators, etc. Note that for $a = 1$ R_A and R_B coincide and $\omega' \big|_{\text{RegA}} = \omega' \big|_{\text{RegB}}$ as can be seen from (2.35).

Fermion determinants can be very easily obtained from (2.18)–(2.19) and (2.35). We list the results: For the TM we have

$$\begin{aligned} \log \det \left[\frac{i\mathcal{D} + gA}{i\mathcal{D}} \right]_{\text{RegIA}} &= \frac{1}{4\pi} (1+a) \int d^2x \, \phi \square \phi \\ &\quad + \frac{i}{4\pi} (a-1) \int d^2x \, \phi \square \eta, \end{aligned} \quad (2.36a)$$

$$\log \det \left[\frac{i\mathcal{D} + gA}{i\mathcal{D}} \right]_{\text{RegIB}} = \frac{1}{4\pi} (1+a) \int d^2x \, \phi \square \phi, \quad (2.36b)$$

$$\begin{aligned} \log \det \left[\frac{i\mathcal{D} + gA}{i\mathcal{D}} \right]_{\text{RegIIA}} &= \frac{1}{4\pi} (1+a) \int d^2x \, \phi \square \phi \\ &\quad + \frac{i}{4\pi} (a-1) \int d^2x \, \phi \square \eta, \end{aligned} \quad (2.36c)$$

$$\log \det \left[\frac{i\mathcal{D} + gA}{i\mathcal{D}} \right]_{\text{RegIIB}} = \frac{1}{4\pi} (1+a) \int d^2x \, \phi \square \phi. \quad (2.36d)$$

For the CSM

$$\begin{aligned} \log \det \left[\frac{i\mathcal{D} + gA((1-\gamma_5)/2)}{i\mathcal{D}} \right]_{\text{RegIA}} &= \frac{1}{8\pi} \int d^2x [(1+a)\phi \square \phi + (a-1)\eta \square \eta - 2i\eta \square \phi], \end{aligned} \quad (2.37a)$$

$$\begin{aligned} \log \det \left[\frac{i\mathcal{D} + gA((1-\gamma_5)/2)}{i\mathcal{D}} \right]_{\text{RegIB}} &= \frac{1}{8\pi} \int d^2x [(1+a)\phi \square \phi - i(1+a)\eta \square \phi], \end{aligned} \quad (2.37b)$$

$$\begin{aligned} \log \det \left[\frac{i\mathcal{D} + gA((1-\gamma_5)/2)}{i\mathcal{D}} \right]_{\text{RegIIA}} &= \frac{(1+a)}{8\pi} \int d^2x [\phi \square \phi - \eta \square \eta - 2i\eta \square \phi], \end{aligned} \quad (2.37c)$$

$$\begin{aligned} \log \det \left[\frac{i\mathcal{D} + gA((1-\gamma_5)/2)}{i\mathcal{D}} \right]_{\text{RegIIB}} &= \frac{(1+a)}{8\pi} \int d^2x [\phi \square \phi - \eta \square \eta - 2i\eta \square \phi]. \end{aligned} \quad (2.37d)$$

Some comments are in order at this point. For the CSM, (2.36a) reproduces the usual result obtained following different approaches.^{1,6,7,10,19–24} Indeed, using the identities

$$\begin{aligned} \partial_\mu A_\mu &= (1/g) \square \eta, \\ \epsilon_{\mu\nu} \partial_\mu A_\nu &= (1/g) \square \phi, \end{aligned} \quad (2.38)$$

Eq. (2.37a) can be rewritten as

$$\log \det \left[\frac{i\hat{D} + gA((1-\gamma_5)/2)}{i\hat{D}} \right]_{\text{RegIA}} = \frac{g^2}{8\pi} \int d^2x A_\mu (\delta_{\mu\alpha} - i\epsilon_{\mu\alpha}) \partial_\alpha \square^{-1} \partial_\beta (\delta_{\beta\nu} + i\epsilon_{\beta\nu}) A_\nu - \frac{ag^2}{8\pi} \int A_\mu^2 d^2x. \quad (2.39)$$

This is, use of the heat-kernel method with $\exp(-\hat{D}_t^2/M^2)$ as regulator and use of the cyclic property of the trace before regularization leads to the result first discussed by Jackiw and Rajaraman.¹ The heat-kernel method with $(DD^+ + D^+D)/2$ as RO does not lead to (2.39). This difference in the results according to whether one uses R_A or R_B is not surprising. As it has been discussed in detail by Fujikawa^{29,30} for non-Abelian anomalies, the choice of R_A or R_B leads to two different forms of the anomaly, namely, the consistent and covariant forms. The same difference occurs using the stochastic quantization method that is analogous to the heat-kernel method either with R_A or R_B (Ref. 36).

It is interesting to note that the extended version of the ζ -function approach presented in Ref. 24 so as to include an a dependence also leads to (2.39).

Introduction of the RO before employing the cyclic property of the trace yields a different result for the determinant

$$\log \det \left[\frac{i\hat{D} + gA((1-\gamma_5)/2)}{i\hat{D}} \right]_{\text{RegII}} \propto \int d^2x A_\mu (\delta_{\mu\alpha} - i\epsilon_{\mu\alpha}) \partial_\alpha \square^{-1} \partial_\beta (\delta_{\beta\nu} + i\epsilon_{\beta\nu}) A_\nu. \quad (2.40)$$

Note that the addition of $\log \det(i\hat{D} + gA_-)$ given by (2.39) and the corresponding result for $\log \det(i\hat{D} + gA_+)$ leads to the Dirac fermion determinant (2.36a) which for $a = 1$ coincides with the well-known gauge invariant Schwinger model determinant.

Concerning the Dirac fermion determinant evaluated for the Thirring model, note that for $a = 1$ (i.e., $R = R_A = R_B = i\hat{D} + gA$), $[R, D] = 0$, hence the four results coincide giving the Schwinger model determinant.

As we mentioned above, it is well known that different RO's lead to different forms of the non-Abelian anomaly. However, physical consequences of these different forms may be identical. In particular, the anomaly cancellation conditions are the same for the covariant and consistent anomalies.³¹ We shall discuss this aspect of our results in the next section.

III. ANOMALIES AND CURRENT-CURRENT COMMUTATORS

From the generating functionals (2.4) and (2.6) we can easily find the vacuum expectation value of fermionic currents:

$$J_\mu^{\text{TM}} = \langle j_\mu(x) \rangle_{\text{TM}} = \frac{1}{\mathcal{Z}_{\text{TM}}} \frac{\delta \mathcal{Z}_{\text{TM}}}{\delta S^\mu(x)} \Big|_{S^\mu=0} = \langle \bar{\psi} \gamma_\mu \psi \rangle, \quad (3.1)$$

$$J_\mu^{\text{CSM}} = \langle j_\mu(x) \rangle_{\text{CSM}} = \frac{1}{\mathcal{Z}_{\text{CSM}}} \frac{\delta \mathcal{Z}_{\text{CSM}}}{\delta A^\mu(x)} = g \left\langle \bar{\psi} \gamma_\mu \left(\frac{1-\gamma_5}{2} \right) \psi \right\rangle. \quad (3.2)$$

[Note that in (3.2) the gauge field A_μ is considered as a background and then it can be used to find J_μ ; for the TM being A_μ an auxiliary field, an external source S_μ has to be introduced in order to compute currents.]

Current-current correlation functions are given by

$$G_{\mu\nu}^{\text{TM}}(x, y) = \langle T(j_\mu(x) j_\nu(y)) \rangle_{\text{TM}} = \frac{1}{\mathcal{Z}_{\text{TM}}} \cdot \frac{\delta^2 \mathcal{Z}_{\text{TM}}}{\delta S^\mu(x) \delta S^\nu(y)} \Big|_{S^\mu=S^\nu=0}, \quad (3.3)$$

$$G_{\mu\nu}^{\text{CSM}}(x, y) = \langle T(j_\mu(x) j_\nu(y)) \rangle_{\text{CSM}} = \frac{1}{\mathcal{Z}_{\text{CSM}}} \cdot \frac{\delta^2 \mathcal{Z}_{\text{CSM}}}{\delta A^\mu(x) \delta A^\nu(y)}, \quad (3.4)$$

and then current-current commutators can be evaluated from (3.3) and (3.4) by means of the Bjorken-Johnson-Low (BJL) limit.^{32,33}

We start with the CSM currents. Relation (3.2) can be rewritten in the form

$$J_\mu^{\text{CSM}} = \frac{\delta}{\delta A^\mu} \log \det D[A], \quad (3.5)$$

and hence the results given in (2.37) for the CSM determinant, written in terms of A_μ can be used. For the regularization leading to the usual form of the fermion determinant [i.e., the regularization RegIA leading to (2.39)] we get

$$J_\mu^{\text{CSM}} = (g/4\pi) \times [(\delta_{\mu\nu} - i\epsilon_{\mu\nu}) \partial_\nu (\phi + i\eta)] - (g^2/4\pi) a A_\mu, \quad (3.6)$$

and hence

$$\partial_\mu J_\mu^{\text{CSM}} = (g^2/4\pi) [(1-a) \partial_\mu A_\mu + i\epsilon_{\mu\nu} \partial_\mu A_\nu].$$

From this well-known result one infers that the theory is anomalous whatever the value of a is chosen. As proved by Jackiw and Rajaraman,¹ the effective action for A_μ resulting from the addition of the fermion determinant (2.39) and the $\frac{1}{4}F_{\mu\nu}^2$ terms defines a sensible unitary Lorentz invariant theory provided $a > 1$ with a massive gauge meson (with mass $g^2[a^2/(a-1)]$ and massless excitations). It is important to stress at this point that any of the three other regularizations lead to this result,¹ confirmed by many other investigations.¹⁹⁻²⁴

Concerning the other regularizations for the CSM, use of R_B instead to R_A leads to

$$J_\mu^{\text{CSM}} \Big|_{\text{RegIB}} = (g^2/8\pi)(1+a)\epsilon_{\mu\nu} A_\nu \quad (3.7)$$

and

$$\partial_\mu J_\mu^{\text{CSM}} \Big|_{\text{RegIB}} = (g^2/8\pi)(1+a)\epsilon_{\mu\nu} \partial_\mu A_\nu, \quad (3.8)$$

which coincides with (3.6) only for $a = 1$. Neither this result nor those arising from the Type II regularization,

$$J_\mu \Big|_{\text{RegII}} = f(a)(A_\mu + \epsilon_{\mu\nu} A_\nu), \quad (3.9)$$

[with $f(a) = (g^2/4\pi)(1+a)$ for types A, B of RO] lead

to the results obtained by many authors using different approaches and reproduced by type IA regularization.

Current-current commutators for the CSM are, as it is well known,³⁴ anomalous (in particular $[j_0, j_0] \propto \delta'$). Only if gauge degrees of freedom are incorporated as dynamical ones as proposed in Refs. 2–5 one gets $[j_0, j_0] = 0$ as proved in Ref. 35.

Concerning the TM we shall compute current-current commutators in order to compare the results obtained using different regularization with the well-known Klaiber results.¹⁶

As we stated above we use the BJL method^{32,33} that starts from the identity

$$\langle [j_\mu(x); j_\nu(y)]_{\text{e.t.}} \rangle = \lim_{\epsilon \rightarrow 0} [G_{\mu\nu}(x_1, x_0; y_1, x_0 + \epsilon) - G_{\mu\nu}(x_1, x_0; y_1, x_0 - \epsilon)] \quad (3.10)$$

and using Eq. (3.3) for evaluating $G_{\mu\nu}^{\text{TM}}$. If we write the fermion determinants listed in (2.36) in the form:

$$\log \det \left[\frac{i\partial + gA}{i\partial} \right] = -\frac{g^2}{2\pi} \int d^2x A_\mu D_{\mu\nu} A_\nu, \quad (3.11)$$

it is easy to see that

$$\langle j_\mu(x) j_\nu(y) \rangle = (1/g^2) [-\delta_{\mu\nu} \delta(x-y) + \mathcal{D}_{\mu\nu}^{-1}(x, y)], \quad (3.12)$$

where

$$\mathcal{D}_{\mu\nu} = \delta_{\mu\nu} + (g^2/\pi) D_{\mu\nu}. \quad (3.13)$$

Only $\square^{-1}$ terms in $\mathcal{D}_{\mu\nu}$ contribute to the $\epsilon \rightarrow 0$ limit in (3.10). Indeed being

$$\square_{xy}^{-1} = (1/2\pi) \log \mu |x - y| \quad (3.14)$$

it appears in $G_{\mu\nu}(x_1, x_0; y_1, y_0 + \epsilon)$ through a term $G_{\mu\nu}^{\square^{-1}}$,

$$G_{\mu\nu}^{\square^{-1}} \propto \frac{\epsilon(x_1 - y_1)}{\epsilon^2 + (x_1 - y_1)^2} \xrightarrow{\epsilon \rightarrow 0} \delta'(x_1 - y_1) \quad (3.15)$$

responsible for a δ' behavior in $[j_0, j_1]$. All other contributions to $G_{\mu\nu}$ vanish when computing the rhs in (3.10).

It is important to stress that if one uses R_B (either with RegI or RegII) one then gets the same results obtained using the operator approach (see, for example, Ref. 16). In particular $[j_0, j_0]_{\text{e.t.}} = 0$ and

$$[j_0; j_1]_{\text{e.t.}} = (1/g^2) h(a) \delta'(x), \quad (3.16)$$

with

$$h(a) = \begin{cases} (1 + \pi/g^2 a)^{-1}, & \text{for RegIB,} \\ (1 + 2\pi/g^2(1+a))^{-1}, & \text{for RegIIB.} \end{cases} \quad (3.17)$$

The a dependence of the coefficient corresponds to the existence of a one-parameter family of solutions in the TM. As stated by Klaiber, this corresponds to an undetermination of the coupling constant in the sense that any value can be produced with an appropriate current definition (or equivalently with an appropriate value of the regularization parameter a).

Concerning regularization A it leads to an anomalous result for $[j_0, j_0]$,

$$[j_0(x); j_0(y)] = k(a) \delta'(x). \quad (3.18)$$

Only for $a = 1$, this choice leads to a sensible result.

IV. SUMMARY AND CONCLUSIONS

We have studied regularization ambiguities in fermionic theories using the path-integral approach. We have chosen as samples the Thirring model and the chiral Schwinger model, two well-known, two-dimensional theories in which gauge invariance cannot be used as a guiding principle when selecting a regularization prescription.

We have employed a heat-kernel regularization, which consists of inserting $\exp(-R/M^2)$ in ill-defined quantities (traces, fermionic Jacobians, etc.) with R some positive definite operator.

While in gauge-invariant theories identification of R with a square of the covariant derivative leads to gauge invariant and unique results (for fermion determinants, chiral anomalies, etc.) in the present models there is a wider choice of regulators, which yield a family of solutions depending (for the two-dimensional case) on an *a priori* arbitrary parameter.

A similar situation is encountered in the study of non-Abelian anomalies^{30,31,36}: depending on the specification of the fermionic generating functional (i.e., the fermionic determinant) two different forms of Ward–Takahashi identities can be obtained. The corresponding anomalies are known as covariant and consistent anomalies and both have important applications in different physical contexts (see, for example, the discussion in Ref. 30). It has been shown by Bardeen and Zumino³¹ that one can always pass from one form of the anomaly to the other one by changing the definition of fermion currents and hence the anomaly cancellation conditions are the same for either form.

Inspired in Fujikawa's discussion³⁰ of these non-Abelian anomalies using the path integral framework, we have tested two forms for RO [Eqs. (2.19a) and (2.19b)] (for non-Abelian anomalies R_A leads to the covariant result and R_B to the consistent one). Moreover, since neither R_A nor R_B commute with the Dirac operator appearing in the Lagrangian, regularization is not fully specified until one decides the step at which the RO is inserted in ill-defined functional traces. Different choices lead to the nonequivalent expressions (2.30) and (2.31) and (2.32) and (2.33).

We can conclude that, as it happens for non-Abelian anomalies, different physical contexts require different choices of regularization when studying chiral models or purely fermionic models (like the Thirring model). For the chiral Schwinger model, analytic continuation of the (originally non-Hermitian) Dirac operator and insertion of the RO after having used the cyclic property of the (ill-defined) trace leads to the definition of a consistent unitary, Lorentz invariant theory, as first discovered in Ref. 1 using a different approach. This result cannot be obtained if one adopts the other regularization schemes we have described.

For the Thirring model, a pure fermionic theory where A_μ is an auxiliary field, insertion of the RO before cyclic property of the trace is used, leads to a sensible answer (for R_B) coinciding with Klaiber operator result.¹⁶

In both cases there remains an ambiguity related to an undetermined parameter (a) introduced through the RO. For the CSM this leads to a whole family of consistent solutions provided $a > 1$ (the resulting theory consists of massless excitations plus a massive particle with mass $(g^2/4\pi)[a^2/(a-1)]$ and hence unitarity requires $a > 1$).

For the Thirring model the existence of a one-parameter family of solutions was already known from the operator analysis described in Ref. 16.

We then conclude that the physical situation must be carefully analyzed in order to select a regularization prescription. Moreover, ambiguities arising in this process of regularization can be utilized in order to define a physically sound theory. Although our discussion corresponds to the (simpler) two-dimensional world, we think that our conclusions also hold in other space-time dimensions. Indeed, different specification of fermion determinants and anomalies are also encountered in four-dimensional non-Abelian models, in gravitational theories in $4k + 2$ dimensions, etc. In this context, it may be worth while to investigate the introduction of arbitrary parameters that can be used to define consistent quantum theories from potentially anomalous ones. We hope to address this point in a future work.

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APPENDIX: DETAILS ON THE CALCULATION OF FERMION DETERMINANTS

In Sec. II C we obtained for the Jacobian associated to the fermionic change of variables (2.10)–(2.12) the result:

$$J = \exp\left(-\int_0^1 \omega'(t) dt\right), \quad (A1)$$

where $\omega'(t)$ takes different forms depending on which regularization is used.

In the first method we used (RegI) we inserted RO after using the cyclic property of the trace and in

$$\begin{aligned} \omega'_{\text{TM}}(t) \big|_{\text{RegI}} &= \lim_{M^2 \rightarrow \infty} 2\text{tr} \gamma_5 \phi e^{-R/M^2}, \\ \omega'_{\text{CSM}}(t) \big|_{\text{RegI}} &= \lim_{M^2 \rightarrow \infty} \text{tr} \gamma_5 (\phi - i\eta) e^{-R/M^2}. \end{aligned} \quad (A2)$$

In the second method (RegII) the cyclic property of the trace was not used. We then obtained

$$\begin{aligned} \omega'_{\text{TM}}(t) \big|_{\text{RegII}} &= \lim_{M^2 \rightarrow \infty} \text{tr} \mathcal{D}_t^{-1} [(\gamma_5 \phi - i\eta) \mathcal{D}_t \\ &\quad + \mathcal{D}_t (\gamma_5 \phi + i\eta)] e^{-R/M^2}, \\ \omega'_{\text{CSM}}(t) \big|_{\text{RegII}} &= \lim_{M^2 \rightarrow \infty} \text{tr} \mathcal{D}_t^{-1} [\tilde{\mathcal{D}}_t ((1 - \gamma_5)/2)(\phi - i\eta) \\ &\quad - ((1 + \gamma_5)/2)(\phi - i\eta) \tilde{\mathcal{D}}_t] e^{-R/M^2}. \end{aligned} \quad (A3)$$

In this Appendix we give some details on the calculation of $\omega'(t)$ leading to the results listed in Eqs. (2.35) and (2.36).

We begin with evaluation of $\omega'_{\text{TM}}(t) \big|_{\text{RegI}}$. Introducing a representation of δ function we have

$$\begin{aligned} \omega'_{\text{TM}}(t) \big|_{\text{RegI}} &= \lim_{\substack{M^2 \rightarrow \infty \\ y \rightarrow x}} \text{tr} \frac{2}{(2\pi)^2} \\ &\quad \times \int d^2x d^2k e^{iky} e^{-D_t^2/M^2} \gamma_5 \phi(x) e^{-ikx} \end{aligned} \quad (A4)$$

and then

$$\begin{aligned} \omega'_{\text{TM}}(t) \big|_{\text{RegI}} &= \lim_{M^2 \rightarrow \infty} \text{tr} \frac{\gamma_5}{2\pi^2} \int \phi(x) dx \\ &\quad \times \int d^2k e^{-(D_t + k)^2/M^2}. \end{aligned} \quad (A5)$$

Expanding the exponential and making the change of variables $k = Mp$, it is easy to see that only the M^{-2} term in the expansion will contribute in the limit M^2 .

Then

$$\omega'_{\text{TM}}(t) \big|_{\text{RegI}} = \frac{1}{2\pi^2} \int d^2x \text{tr} \gamma_5 \phi(x) D_t^2 \int e^{-p^2} d^2p. \quad (A6)$$

Performing the Gaussian integral and using the corresponding RO we finally obtain the result given in (2.35a).

Concerning RegIB note that although R_B cannot be written as the square of some operator, one can show the fact that calculations are equivalent to those performed with an $R_B = \mathcal{D}_t^2$, with

$$\mathcal{D}_t = i\mathcal{D} + g((1+a)/2)\mathcal{A}_t. \quad (A7)$$

Now let us consider the second method (RegII). In this case we have

$$\begin{aligned} \omega'_{\text{TM}}(t) \big|_{\text{RegII}} &= \lim_{M^2 \rightarrow \infty} \text{tr} D_t^{-1} [(\gamma_5 \phi - i\eta) \mathcal{D}_t \\ &\quad + \mathcal{D}_t (\gamma_5 \phi + i\eta)] e^{-R/M^2}. \end{aligned} \quad (A8)$$

The trace in (A8) implies one has to consider $\lim_{x \rightarrow y} D_t^{-1}(x, y)$. It is this divergent contribution that is the new ingredient in RegII that leads to a different result compared with RegI.

Inserting

$$\begin{aligned} D_t^{-1}(x, y) &= e^{[\gamma_5 \phi(x) + i\eta(x)](1-t)} \\ &\quad \times G_0(x, y) e^{[\gamma_5 \phi(y) - i\eta(y)](1-t)} \end{aligned} \quad (A9)$$

with G_0 , the free fermion Green's function,

$$G_0(x) = (i/2\pi)(\gamma_\mu x_\mu/x^2) \quad (A10)$$

in (A8) one gets

$$\begin{aligned} \omega'_{\text{TM}}(t) \big|_{\text{RegII}} &= \lim_{M^2 \rightarrow \infty} \int d^2x d^2k \frac{1}{2\pi} \text{tr} \\ &\quad \times \left[\frac{k}{k^2} i\mathcal{D}(\gamma_5 \phi + i\eta) \right] e^{-ik(x-y)}. \end{aligned} \quad (A11)$$

Now, following the same steps for the RegI case we get Eqs. (2.36).

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